

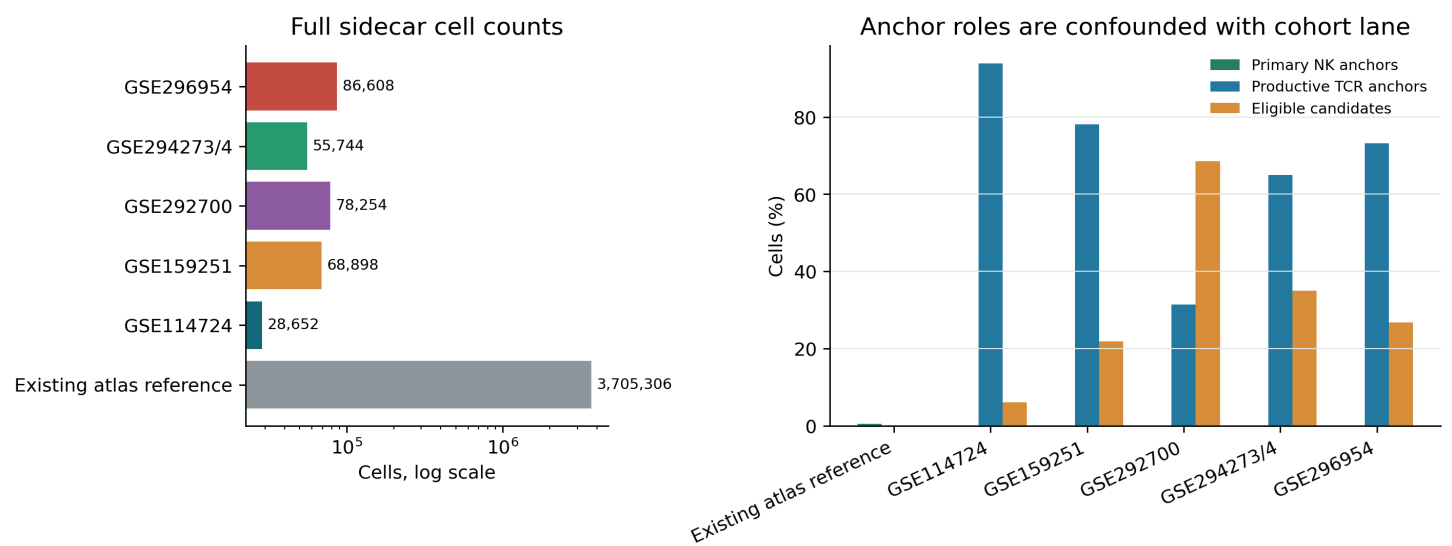
gdTAI V4.2 development sidecar diagnostics

This is a diagnostic visualization of the development-only scVI sidecar. It is not a new canonical atlas milestone and contains no locked validation cohort. Full-data counts use all 4,023,462 cells; UMAP and local-neighbor plots use a deterministic, source-balanced 250,000-cell sample.

Interpretation: the integration contains an NK-anchor-rich region, but the current method cannot confidently extend that region to unlabeled NK cells. Source balance, boundary localization, and anchor/cohort confounding all prevent a defensible pseudo-label set.



What was integrated



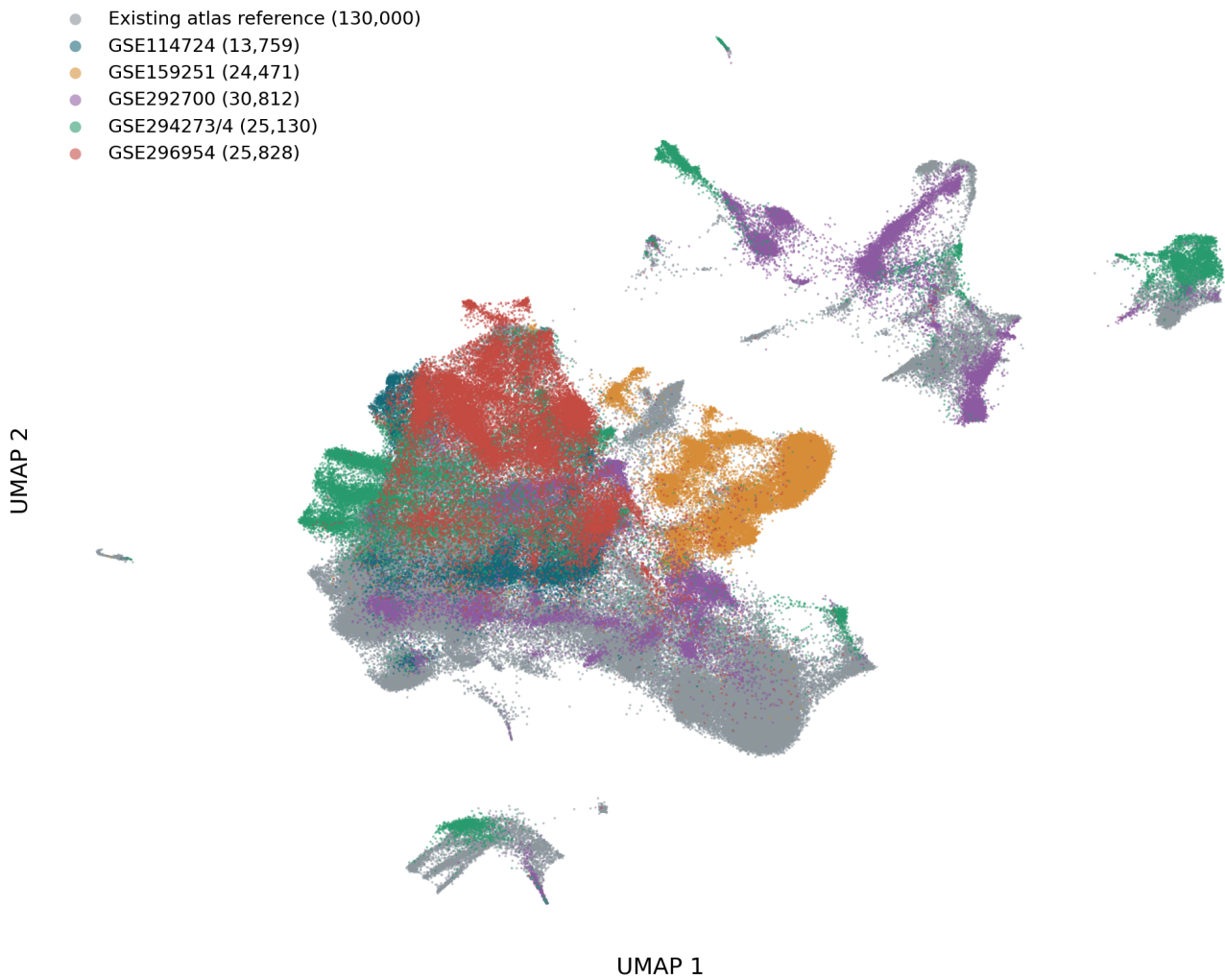
Full-data cohort sizes and evidence classes. Candidate eligibility is intentionally restricted to the five new development cohorts.

Cohort	Cells	Primary NK	Primary NK %	Productive TCR	Productive TCR %	Eligible candidates	Candidate %
Existing atlas reference	3705306	21054	0.57%	0	0.00%	0	0.00%
GSE114724	28652	0	0.00%	26893	93.86%	1759	6.14%
GSE159251	68898	0	0.00%	53773	78.05%	15125	21.95%
GSE292700	78254	0	0.00%	24577	31.41%	53677	68.59%
GSE294273/4	55744	0	0.00%	36221	64.98%	19523	35.02%
GSE296954	86608	0	0.00%	63405	73.21%	23203	26.79%

Critical confounding: all 21,054 primary NK anchors are from the existing atlas reference, whereas all 204,869 productive-TCR anchors are from the five new cohorts. The clustering cannot cleanly distinguish lineage structure from old-versus-new cohort structure.

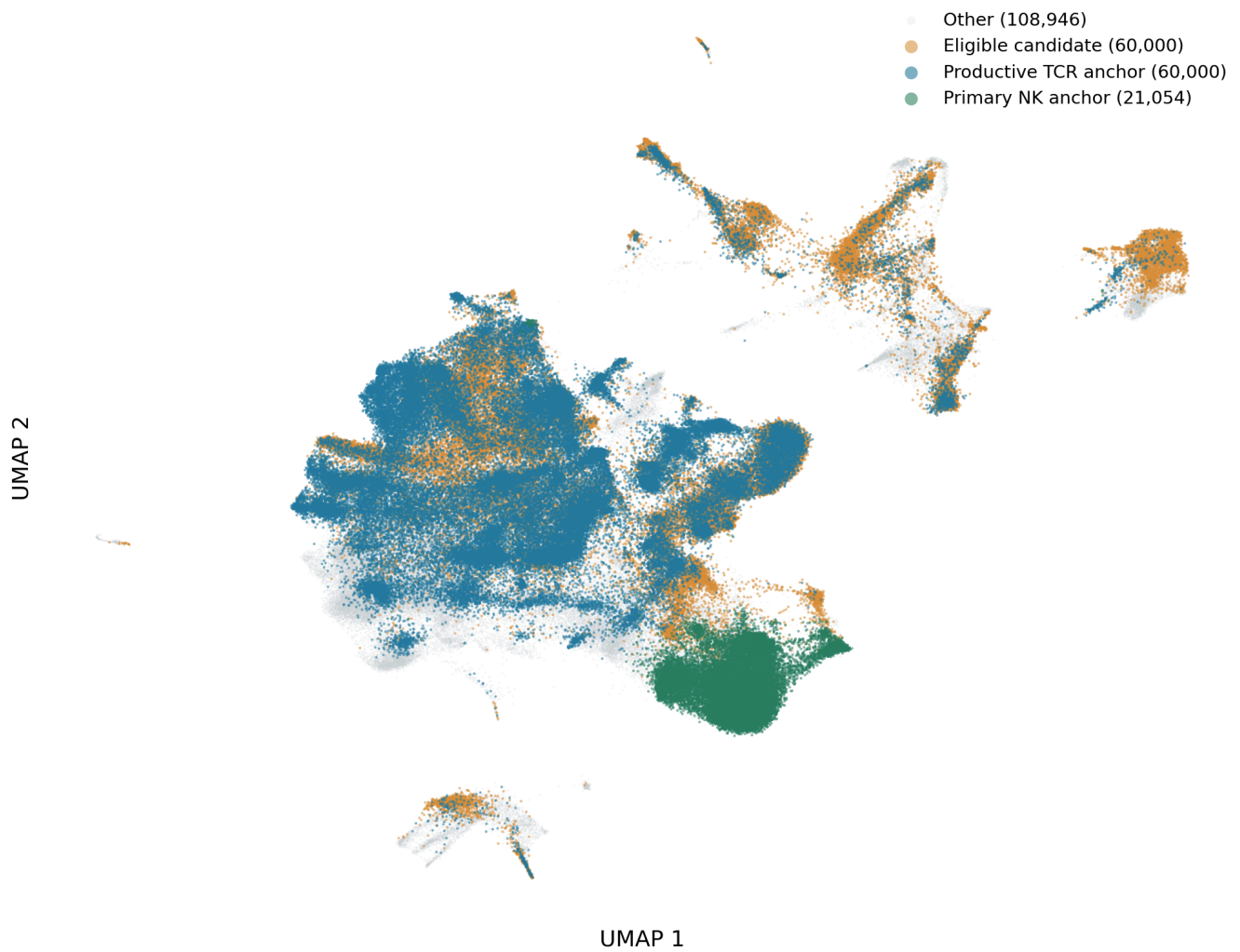
Latent-space structure

V4.2 development sidecar by input cohort



Diagnostic scVI UMAP by input cohort.

Anchor classes and candidates (anchor class is cohort-confounded)

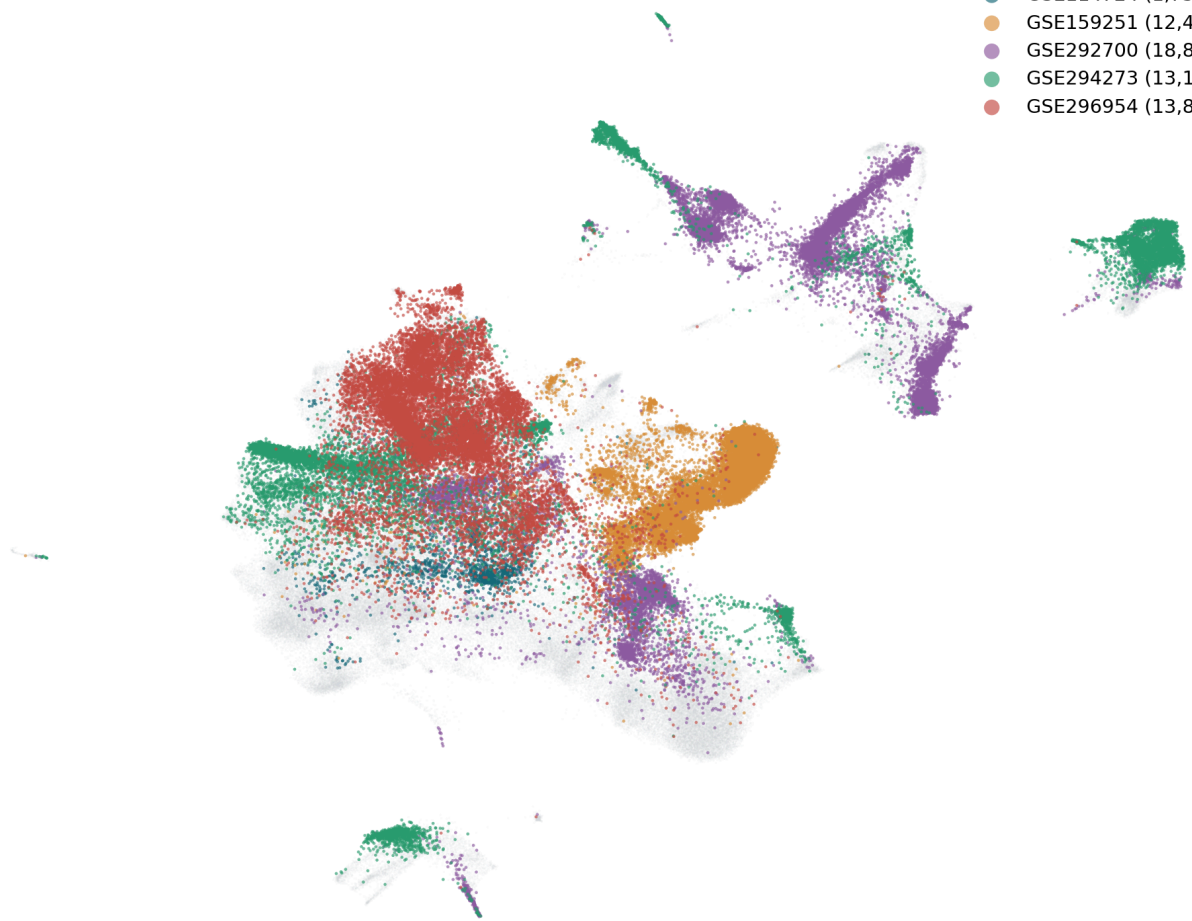


Primary NK and productive-TCR anchors are expression-independent; candidates are no-TCR, non-doublet cells from permitted development cohorts.

Eligible pseudo-NK candidates by source

- GSE114724 (1,759)
- GSE159251 (12,471)
- GSE292700 (18,812)
- GSE294273 (13,130)
- GSE296954 (13,828)

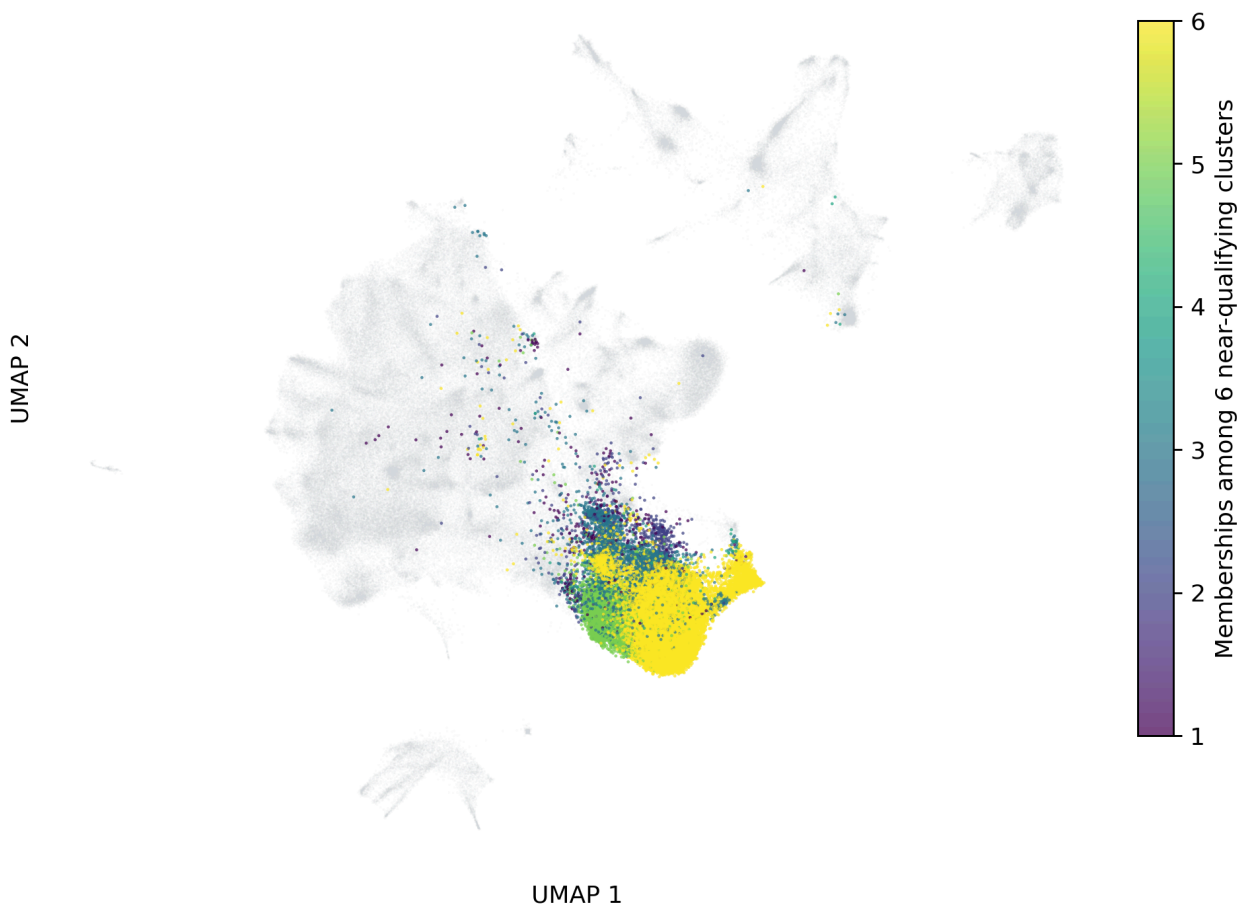
UMAP 2



UMAP 1

Eligible candidates show strong source-dependent localization.

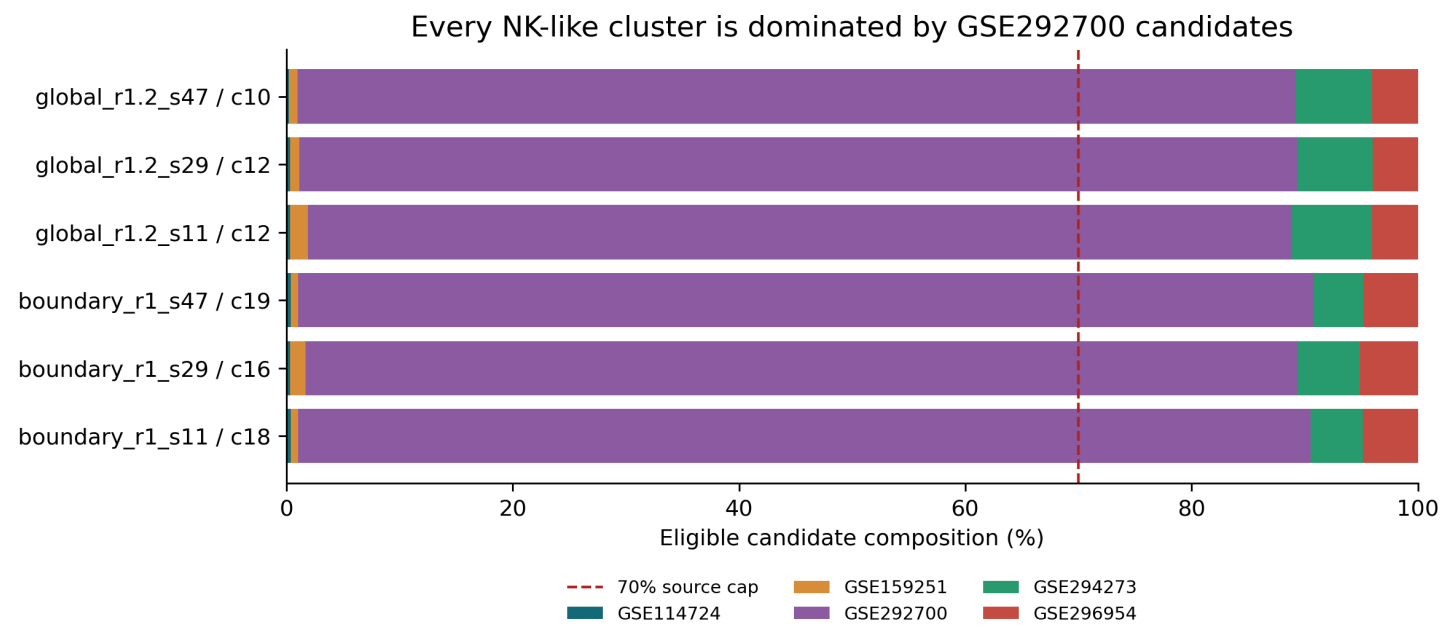
Stability of the anchor-defined NK-like region



Membership frequency across the six clusters that passed anchor purity and TCR contamination criteria.

Why pseudo-NK selection failed

All six NK-anchor-rich clusters were low in productive-TCR contamination, but their eligible candidate cells were 86.92%-89.69% GSE292700. This violates the frozen 70% maximum-source rule, so no cluster qualified and no candidate cell received a valid consensus label. Because the two anchor classes come from disjoint cohort lanes, anchor purity alone cannot prove lineage identity.



The dashed line is the frozen 70% maximum contribution from one candidate source.

The NK-anchor region is stable but cohort-confounded

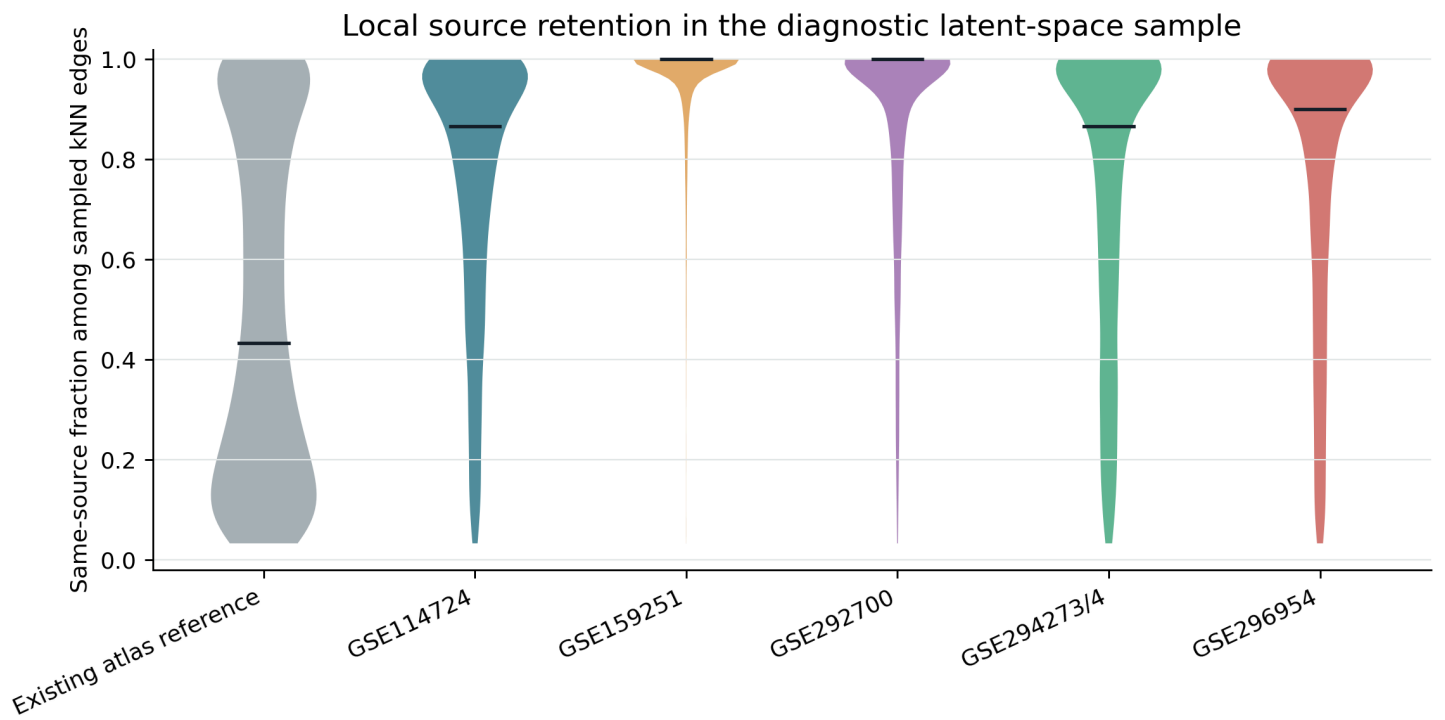


The anchor-defined region is stable across runs, but cohort-confounded anchors and source-skewed candidates prevent a defensible pseudo-label set.

Run	Cluster	NK anchors	TCR anchors	NK purity	TCR/cell	Dominant source	Source share
global_r1.2_s11	12	19755	328	98.37%	0.10%	GSE292700	86.92%
global_r1.2_s29	12	19751	308	98.46%	0.10%	GSE292700	88.17%
global_r1.2_s47	10	17221	263	98.50%	0.12%	GSE292700	88.19%
boundary_r1_s11	18	20267	799	96.21%	0.20%	GSE292700	89.46%
boundary_r1_s29	16	20267	801	96.20%	0.20%	GSE292700	87.65%
boundary_r1_s47	19	20113	813	96.11%	0.21%	GSE292700	89.69%

Integration mixing diagnostic

The same-source-neighbor fraction is a descriptive diagnostic, not a biological pass/fail metric: true tissue, disease, and phenotype differences can legitimately retain source structure. It nevertheless shows where new cohorts remain locally isolated after scVI.



Same-GSE fraction among sampled 30-nearest-neighbor graph edges.

Cohort	Sample cells	Median same-source	Mean same-source	90th percentile
Existing atlas reference	130000	46.67%	51.01%	100.00%
GSE114724	13759	86.67%	75.32%	100.00%
GSE159251	24471	100.00%	95.64%	100.00%
GSE292700	30812	100.00%	87.10%	100.00%
GSE294273/4	25130	86.67%	73.89%	100.00%
GSE296954	25828	90.00%	76.76%	100.00%

Conclusion

An NK-anchor-enriched region is visible, but it cannot yet establish NK identity for unlabeled cells. The next iteration must first break the anchor/cohort confounding, then use source-balanced candidate sampling and local anchor-neighborhood propagation with explicit rejection of source-specific neighborhoods. It should not loosen the source cap based on these results.

Classifier fitting performed: **no**. Source H5AD mutation: **none**. GitHub publication: **disabled until V4 is finished**.

Reproducibility

- Latent SHA-256: 7030b28be885b85a4b24ce2223bb16f7c7288a29195f2512bf7bbba330791ccf
- Partitions SHA-256: 90e83ec83986358a015885e02e29d06eacf726d928e02b2e90428d5c09947a63
- Diagnostic sample seed: 20260816
- Report script SHA-256: a9548ad42e64437334a04cccec8c85d1a2f42c58a10f32a5d9dfa3c223e852b5c